V4 wiring layout (applied to the CAD)

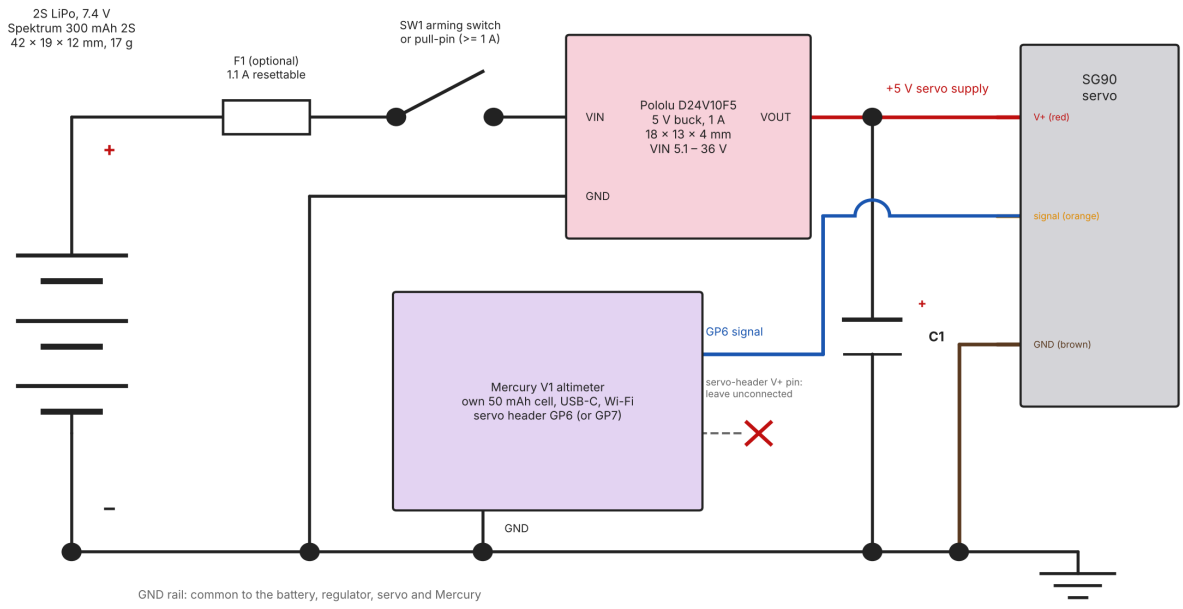
Where each wire runs inside and around the rebuilt V4 model, drawn on sections of the CAD. The Mercury V1 photo you sent gave the header pad order, and the wire paths below are now cut into the model. The electrical schematic is unchanged (see the power and wiring document).

What the check found, and what was done: the horn closes the top of the rotor bore, so the servo cable cannot go down the hollow rotor. The cable now runs in an **outer groove** on the 135° strip and through a hole into the bay. To give the Mercury's soldered wires headroom the bay was made **2.0 mm longer** (housing 96.3 mm). **Pad positions are estimated from a photo** (6 pads at about 2.54 mm pitch, GND at x –3.5 mm with the button end at –x): check them on the real board before printing.

1 Mercury header and harness

From your photo, the pads along the right-hand long edge are, from the button end: **GND, GP6, GP7, OUT, IN, 3V**. The servo needs only **GND** and **GP6** (adjacent pads); the servo's +5 V comes from the regulator, not from the 3V pad. The Mercury sits with this edge toward the centre wall of the sled and the button end at –x, so GND and GP6 are the pads closest to the end where the servo cable arrives.

V4 airbrake: servo power and signal (wiring schematic)



C1 = 470 - 1000 uF, 10 V, low-ESR, wired right at the servo pins. The regulator powers the servo; the Mercury only sends the position signal.

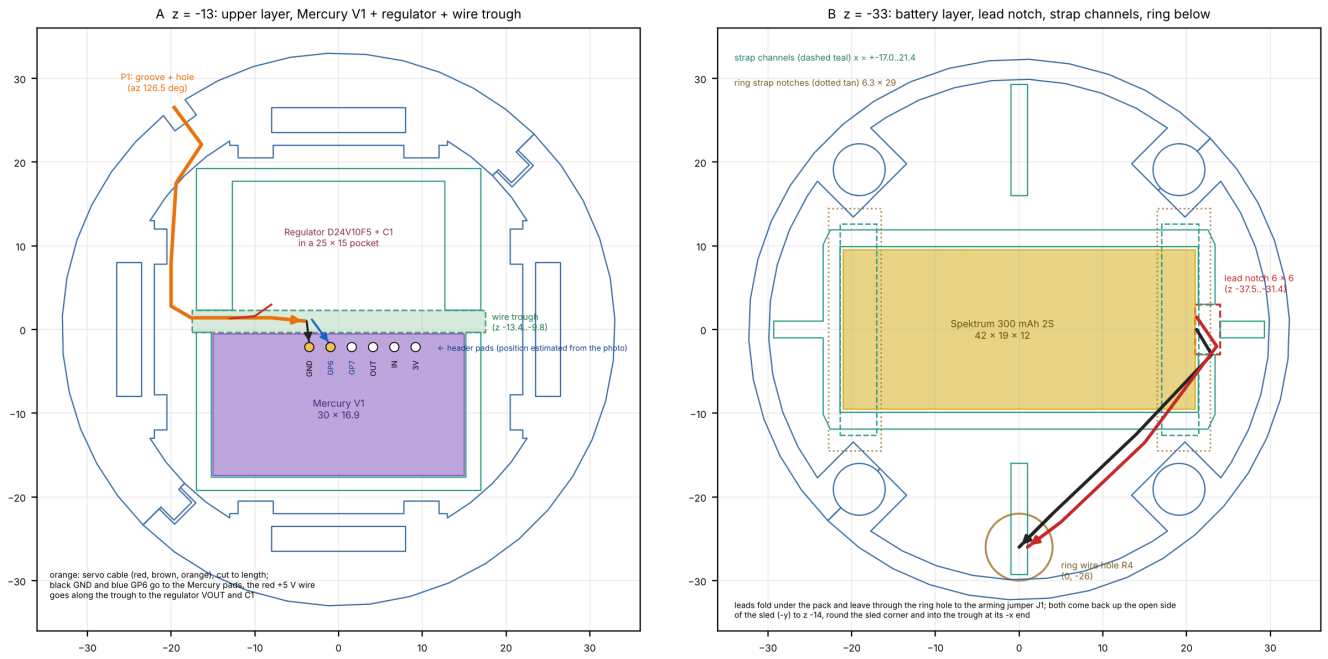
Ground is shared. Never join the servo red wire to a Mercury pin; confirm the Mercury servo-header pin order in its manual before wiring.

Reference schematic (battery label now Spektrum 300 mAh 2S).

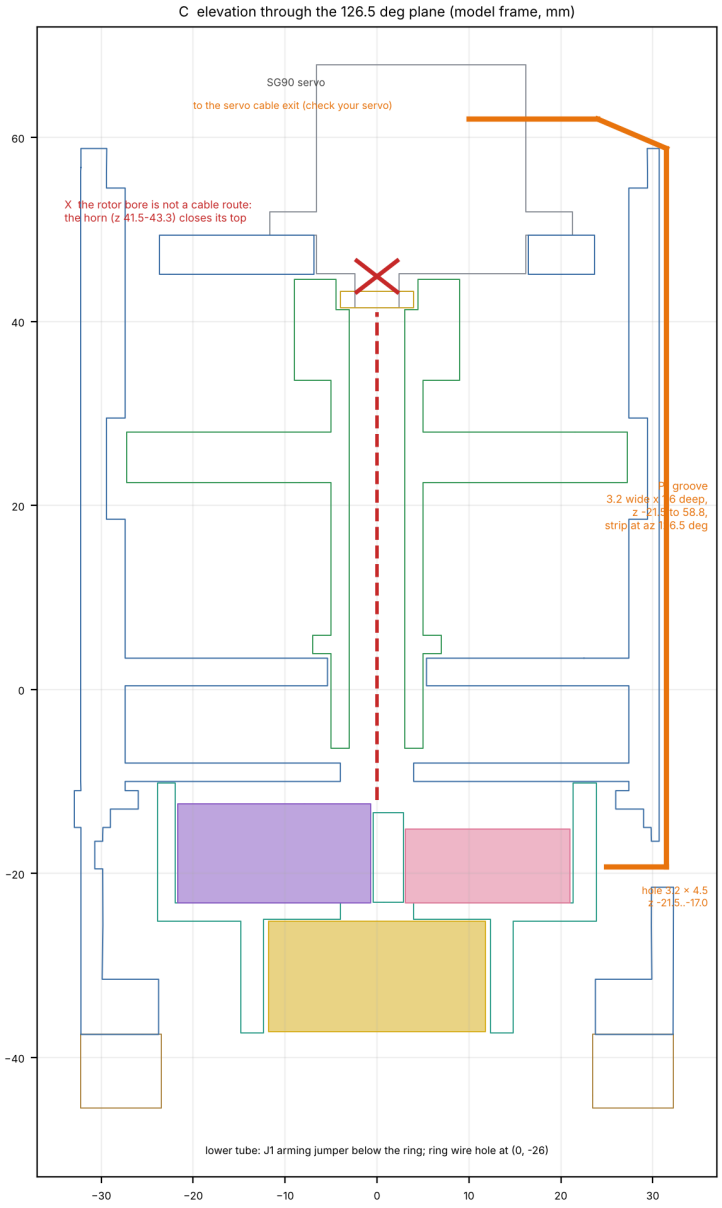
Wire	Function	From → to	Wire	Length (est.)
W1	Battery + through the arming jumper	pack plug → polyfuse → J1 (lower tube) → regulator VIN	24 AWG red	60 mm + 110 mm
W2	Battery –	pack plug → regulator GND (the ground rail)	24 AWG black	105 mm
W3	+5 V	regulator VOUT → C1+ → servo red (inside the servo cable)	24 AWG red	40 mm + cable
W4	Ground rail	regulator GND → C1– → Mercury GND pad → servo brown	24 AWG black	40 mm + cable
W5	Servo cable (3 cores)	servo → groove (z 58.8 → –21.5) → hole → bay → trough	servo's own lead	about 190 mm
W6	Signal	Mercury GP6 pad → servo orange	28 AWG	25 mm
J1	Arming jumper	remove-before-flight 2-pin plug in the lower tube, just below the ring wire hole	JST-PH or header	—

Lengths are read off the drawn routes (±15 mm) without slack: add about 20 mm each. The servo's own lead is usually 200–250 mm (seller dependent): cut it at the bay end. Cap C1 (470–1000 µF, 6.3 × 11 mm) stands beside the regulator; the pocket is now 13.3 mm deep.

2 Layout drawings



Plan sections of the rebuilt model with the routes (orange: servo cable, red: +5 V and battery +, black: battery –, blue: GP6). The trough, lead notch, strap channels and ring notches are cut in the model.



Elevation through the 126.5° plane. Orange: the servo-cable route (groove and hole). Red dashed with a cross: the rotor bore, which does not work as a cable route.

3 What was cut into the V4 model

Item	Problem	Change in the model	Touches
Bay 2.0 mm longer	Wires soldered to the Mercury's through-hole pads need 2 mm of headroom; the Mercury top was 0.4 mm below the ceiling.	Battery layer and everything below moved down 2.0 mm; the Mercury and regulator pockets lowered 2.0 mm (floor between the layers stays 1.8 mm). Housing 94.3 → 96.3 mm (z –37.5 … 58.8), transition ring z –45.5 … –37.5.	housing, sled, ring
P1 cable groove + hole	Horn blocks the rotor bore; the wall is too thin for a hole; the 0.7 mm gap under the upper tube is too small.	Outer groove 3.2 wide × 1.6 deep at azimuth 126.5° on the 135° strip, z –21.5 … 58.8; hole 3.2 × 4.5 into the bay at z –21.5 … –17.0. The strips are only 25° wide between the 65° fins, and the 135°/315° strips carry no seam screws. A 3 mm ribbon fits the groove and the hole without touching any part (checked).	housing halves
P2 wire trough	Pocket tops were 0.2 mm under the ceiling: no way in.	Trough along the centre wall of the sled, y –0.35 … 2.3, x ±17.5, z –13.4 … –9.8; it joins the Mercury and regulator pockets and is open at both ends. A route from the groove hole round the sled corner to the trough keeps at least 0.8 mm from every part (checked).	sled
P3 lead notch	Battery leads had 0.4 mm at the pack end.	Notch 6 wide × 6 tall in the +x pocket end wall (y ±3, z –37.5 … –31.4), open to the bottom.	sled
P5 vents	Mercury reads static pressure.	Three 3 × 3 mm holes through the lower sleeve at z –18.5 … –15.5, azimuths 0°, 120°, 240°, and three O3 holes in the reference lower tube.	housing, tube

P4 (a riser groove on the sled) is no longer needed: the battery leads come up the open side of the sled to z –14 and round the sled corner into the trough.

4 Harness assembly order

- Bench-test first: servo, regulator and Mercury on a table with a 5 V supply; confirm the servo moves from the Mercury signal alone (power only from the regulator).
- Prepare the servo lead: pull the red wire out of the Mercury side and keep it for the +5 V line; solder 28 AWG leads from the brown and orange to the Mercury GND and GP6 pads, from the top, bending them flat toward the centre wall.
- Fit the servo to the ledges (M2×6, O1.8 pilots), feed the cable down the outer groove, through the hole, round the sled corner and into the trough.
- Fill the sled: regulator and C1 in their pocket, Mercury in its pocket (header edge to the trough), wires in the trough; battery in the bottom pocket with its leads folded out through the notch; strap it (two ties or one hook-and-loop strap).
- Run battery + and – down through the ring wire hole to J1 in the lower tube and back up; leave J1 open (disarmed).
- Slide the sled up, fit the transition ring, check that no wire is pinched between the sled and the ring (the ring has strap notches), and only then close the jumper.

5 Still open

- The pad positions come from a photo: confirm the header position and pad pitch on the real Mercury (and its case: it has a loop tab) before printing the sled.
- Where the servo cable leaves your SG90 (the drawing assumes the end that is on top) and its length.
- The Spektrum pack's plug type (PH 2.0 listed) and its lead length.
- Not printed or tried: groove depth against your wire, and whether the cable bends easily at the sled corner.